Bhanu Agarwal

M.Sc. NLP Student | Python, SQL, Semantic Search

Trier, Germany

bhanuagarwal841@gmail.com

GitHub: bhanu613 | LinkedIn: agarwalbhanu

Education

04/2025 – present **M.Sc. Natural Language Processing**, University of Trier, Germany
Current average: 1.7 across 50 ECTS of passed examinations.

10/2023 – 03/2025 **M.Sc. Data Science**, University of Erlangen-Nürnberg (FAU), Germany
Transferred to Trier to specialise in Natural Language Processing.

07/2019 – 05/2023 **B.Tech. Computer Science and Engineering**, Vellore Institute of Technology, India
Final grade: 8.22/10

Work Experience

01/2023 – 03/2023 **Software Development Intern**, Capella Infotech Solutions, Chennai, India

- Fixed software bugs and implemented logic specified by senior developers.
- Labelled and segmented customer reviews for a sentiment-classification project.

Selected Projects

JobFit Navigator · jobfit-navigator | Live demo
Python, Streamlit, SQLite, BM25, Sentence Transformers

- Built a Streamlit and SQLite workspace for job listings, profile evidence and source-linked review of supported English and German requirement wording.
- Combined BM25 and pretrained multilingual embedding search over overlapping passages using reciprocal rank fusion (k=60), with metadata filtering before ranking.

Technical Skill Extraction from Job Descriptions · alias-aware-technical-skill-extraction
Python, regular expressions, evaluation

- Built strict and alias-aware lexicon matchers for technical skills in English IT job descriptions.
- Used a reproducible 200/100 split, a frozen 20-label lexicon and manually annotated gold labels; raised micro-F1 from 0.862 to 0.984, recovering 80 of 92 baseline false negatives.

U.S. Amazon Customer Value Analytics · U.S.-Amazon-customer-analytics
Python, pandas, scikit-learn, Matplotlib, Seaborn

- Analysed Amazon purchase and survey data in a university group project; contributed charts, dashboard support and business insights.
- Collaborated on customer-value classification; the team's tuned Gradient Boosting model reached 0.920 ROC-AUC and 0.740 F1 on a held-out test set.

Technical Skills

Programming and tools: Python; SQL; Git/GitHub; Streamlit, SQLite, Docker

Retrieval and NLP: BM25, pretrained multilingual sentence embeddings, semantic and hybrid search; text preprocessing, lexicon design; Hugging Face Transformers (basic)

Data and ML: pandas, NumPy, scikit-learn, PyTorch; precision/recall/F1 and ROC-AUC; Matplotlib, Seaborn, Excel

Languages & Availability

Languages English: C1 | German: A1

Availability Seeking internship and working-student opportunities across Germany.